

Software Reverse Engineering Course

- Site: OnHexGroup.ir
- Youtube: @onhexgroup
- Telegram: onhex_ir
- Twitter: @onhexgroup
- Github: onhexgroup

Windows
(x86 / x64)

By: OnhexGroup

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Number Systems

- How numbers are represented
- Value of each digit
- Which digits are used to represent numbers
- Example:
 - 1402

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The most important number system

- Binary or Base-2
- Decimal or Base-10
- Hexadecimal or Base-16

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Decimal

- The most widely used number system
- Uses digits from 0 to 9
- For Example: 85, 1402, 1234567890

Youtube: Onhexgroup

Binary

- Used in computer systems
- Uses only 0 and 1
- Examples of binary equivalents:
 - $85 = 01010101$
 - $1489 = 010111010001$

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Hexadecimal

- Used to shorten binary representations
- Covers values 0 to 15 (digits 0–9 and letters A–F)
- Example:
 - $85 = 55$
 - $1489 = 5D1$

0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
0	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F

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Key Notes About Number Systems

- Types of number systems
- How number systems are represented:
 - By default
 - 0x123, 123, 123h, 1010b
- Converting between different bases:
 - Windows Calculator
 - Online conversion tools [[Example](#)]
 - Reverse engineering tools